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220751

5th Sem/ Civil
Subject : RCC Design & Drawing

Time : 6 Hrs.

M.M. : 120

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 There are _____ number of grades in concrete.
Q.2 Ordinary Portland cement is available in _____ grades.
Q.3 The Portion of the beam consisting of slab is called as _____.
Q.4 The distribution steel is also called as _____ reinforcement.
Q.5 For slabs spanning in two directions, the _____ of the two spans shall be used for calculating span to effective depth ratios.
Q.6 When the length of slab is _____ the twice its width it is known as two way slab.

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 A footing that supports a single column is known as _____ footing.
Q.8 Buckling in long column is _____.
Q.9 Water-cement ratio of concrete used in pre-stressing should be about _____.

State True or false:

- Q.10 High yield Strength deformed (H.Y.S.D.) bars confirm to IS : 1786. (True/False)
- Q.11 Fe-500 D has better percentage elongation than Fe-415 grade. (True/False)
- Q.12 Cracks which are at 45 degree with the axis of beam are called flexural cracks. (True/False)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Enlist any five difference between mild steel and HYSD steel.
- Q.14 Define the permissible stresses in steel and concrete & Design value for Material & load.
- Q.15 Describe any five difference b/w W.S.M. and L.S.M.
- Q.16 Write down assumptions made in limit state of collapse in compression.
- Q.17 Enlist any five difference between one way slab & two way slab.
- Q.18 Difference between pre-tensioning and post-tensioning methods? (any five)
- Q.19 Describe the term : Pre-stressed concrete also enlist grade used in prestressed concrete.
- Q.20 What is the necessity of providing T-Beam.
- Q.21 A Singly reinforced beam of size 250mm X 450 mm (overall) is reinforced with steel bars of area 1300 mm². Use M20 grade of concrete and Fe-415 steel. Calculate the ultimate moment of resistance of the beam section.
- Q.22 Enlist Various Losses in Pre-Stress Concrete.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Design the reinforcement details of a simply supported slab resting on masonry walls to the following requirements:

a) Clear span = 3.0 m b) Live load = 2.5 kN/m^2

Q.24 Write design steps for two way R.C.C. Slab.

Q.25 Write down the steps for the design of singly reinforced beam using the limit state method design.

SECTION-E

Note: Attempt any three questions out of four questions. (3x20=60)

Q.26 Draw the plan and sectional elevation of a two way slab from the following data:

Size of the room = 5m x 4m

Bearing on walls : 150mm

Thickness of the wall = 300 mm

Thickness of the slab = 150mm

Reinforcement along both the span = 10mm dia bars @ 120 mm C/C with alternate bars bent up at 1/7.

Q.27 Draw to a suitable scale L-Section & a cross-section of a simply supported R.C.C. beam from the following data

Size of beam = 300x500

Clear span = 3000 mm

Bearing of walls = 250 mm

Main Reinforcement = 4-20 mmØ

Side face reinforcement = 2-12 mmØ on each face

Anchor bars = 2-12 mmØ

Shear Stirrups = 8mmØ 2 legged @ 200mm c/c

- B) Explain i) Nominal cover to reinforcement
ii) Anchoring Bars in tension
iii) Calculations of length of steel bars for beams/slabs.

Q.28 Draw the sectional plan and elevation of a square column from the following data :

Column size = 400 mm x 400 mm

Depth below G. 1 = 1m

Column reinforcement :

Longitudinal reinforcement = 4 bars of 25 mm dia

Lateral ties = 8mm dia @ 300 mm c/c

Footing Detail :

Size of footing = 3m x 3m

Reinforcement = 12mm dia @ 299 mm c/c both ways

Thickness of footing = 300mm at free end, 600mm at column face.

Q.29 Draw to a suitable scale cross-section of an end connection of a column with a beams from the following data :

End Column:

Size of column = 300 x 300mm

Main reinforcement = 8-16 mm dia

Lateral ties = 8mm dia @200 mm c/c (double ties)

Beams :

Size of beams = 200mm x 300mm) including 100 mm thick slab)

Main bars = 5-20 mm dia (out of which two bars are bent-up at $L/7$ from centre of column)

Vertical stirrups = 8mm dia 2 legged stirrups @ 200 c/c

Anchor bars = 2-12 mm dia